DocuLens AI

Conversational AI for Documentary Videos

Edwin Alexis Santiago Rodríguez

The Problem

Documentaries are full of knowledge — but finding specific insights means watching hours of footage.

Problem

Valuable content locked in long-form video

Solution

AI chatbot that answers questions from transcripts

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Why it matters

Instant, grounded answers with source timestamps

Technical Summary

Data Source

YouTube Transcript API — per-snippet text with timestamps

Embedding Model

OpenAI Embeddings

Vector Database

Chroma (persisted locally)

LLM

GPT-4o-mini via OpenAI API

Agent Framework

LangChain + LangGraph (3 tools)

Interface

Streamlit + Whisper API (voice input)

Monitoring

LangSmith

Deployment

Streamlit Cloud

Dataset & Data Pipeline

1

Source

YouTube videos
with CC captions

2

Fetch

YouTube Transcript API
per-snippet +
timestamps

3

Clean

Regex noise removal
whitespace
normalization

4

Chunk

RecursiveCharacterTex
tSplitter
size=500, overlap=50

5

Store

Chroma vectorstore
video_id, title,
timestamp

Q&A Agent

Architecture: LangGraph agent with 3 tools

Memory: Full conversation history on each invoke

Prompt: 13 grounding rules — context-only, confidence scoring

Filtering: Score threshold 0.55 — weak matches discarded

Agent Tools

`rag_answer`

Retrieves chunks, generates grounded answer + confidence score

`retrieve_sources`

Returns titles, timestamps & YouTube deep-links

`video_info`

Lists all loaded documentaries from Chroma metadata

Tools & Components

Chroma

Persisted vector database
Add / delete / clear per video

Semantic Search

OpenAI Embeddings
Meaning-based retrieval

GPT-4o-mini

Answer generation
via OpenAI API

Whisper API

Voice input (STT)
Recorded via Streamlit

LangSmith

Production tracing
Run inspection + cost

Streamlit Cloud

Live deployment
Persistent vectorstore

Evaluation — LangSmith

31	0%	2.54s	$0.01
Total Runs	Error Rate	Median Latency	Total Cost

Improvements Made

Score filtering (threshold 0.55) — discards weak matches to reduce hallucinations

Adaptive retrieval k=10 for comparisons — better multi-documentary answers

Confidence scoring (High / Medium / Low) — adds transparency to each response

LangSmith — Live Trace View

youtube-documentary-chatbot ID

Runs | Threads | Evaluators | Automations | Insights

Retention 14d

Default View | Last 1 day | Traces | Runs | Add filter

Reset | Save View

Load latest

Name	Input	Output	Error	Start Time	Latency	Dataset	Annotation Queue	Tokens	Cost	First Toke
retrieve_sourc	What challenges do elephants face in the wild?	Source 1 Title: Giants of the Earth: ...		3/12/2026, 7:...	1.69s			0		
ChatOpenAI	tool: Elephants face several significant challe...	ai: Elephants face several significa...		3/12/2026, 7:...	2.23s			597	$0.0001566	
model	tool: Elephants face several significant challe...	[{"goto":[],"graph":null,"resume":n...		3/12/2026, 7:...	2.24s			597	$0.0001566	
ChatOpenAI	human: You are a documentary question-ans...	ai: Elephants face several significa...		3/12/2026, 7:...	2.58s			1,056	$0.0002097	
rag_answer	{'question': 'What challenges do elephants fa...	tool: Elephants face several signific...		3/12/2026, 7:...	4.52s			1,056	$0.0002097	
tools	tool_call_with_context	tool: Elephants face several signific...		3/12/2026, 7:...	4.52s			1,056	$0.0002097	
ChatOpenAI	human: What challenges do elephants face in...	ai: {"refusal":null}		3/12/2026, 7:...	1.84s			323	$0.0000588	
model	human: What challenges do elephants face in...	[{"goto":[],"graph":null,"resume":n...		3/12/2026, 7:...	1.85s			323	$0.0000588	
LangGraph	user: What challenges do elephants face in th...	ai: Elephants face several significa...		3/12/2026, 7:...	8.74s			1,976	$0.0004251	
retrieve_sourc	where do elephants live	Source 1 Title: Wild India - Fantasti...		3/11/2026, 9:...	1.71s			0		
ChatOpenAI	tool: Elephants live in a variety of habitats acr...	ai: Elephants live in a variety of hab...		3/11/2026, 9:...	2.54s			752	$0.0001578	
model	tool: Elephants live in a variety of habitats acr...	[{"goto":[],"graph":null,"resume":n...		3/11/2026, 9:...	2.55s			752	$0.0001578	
ChatOpenAI	human: You are a documentary question-ans...	ai: Elephants live in a variety of hab...		3/11/2026, 9:...	3.28s			1,055	$0.0002118	
rag_answer	{'question': 'where do elephants live'}	tool: Elephants live in a variety of h...		3/11/2026, 9:...	4.91s			1,055	$0.0002118	

Conclusion

End-to-end RAG system that turns any YouTube documentary into a conversational knowledge base.

What it demonstrates

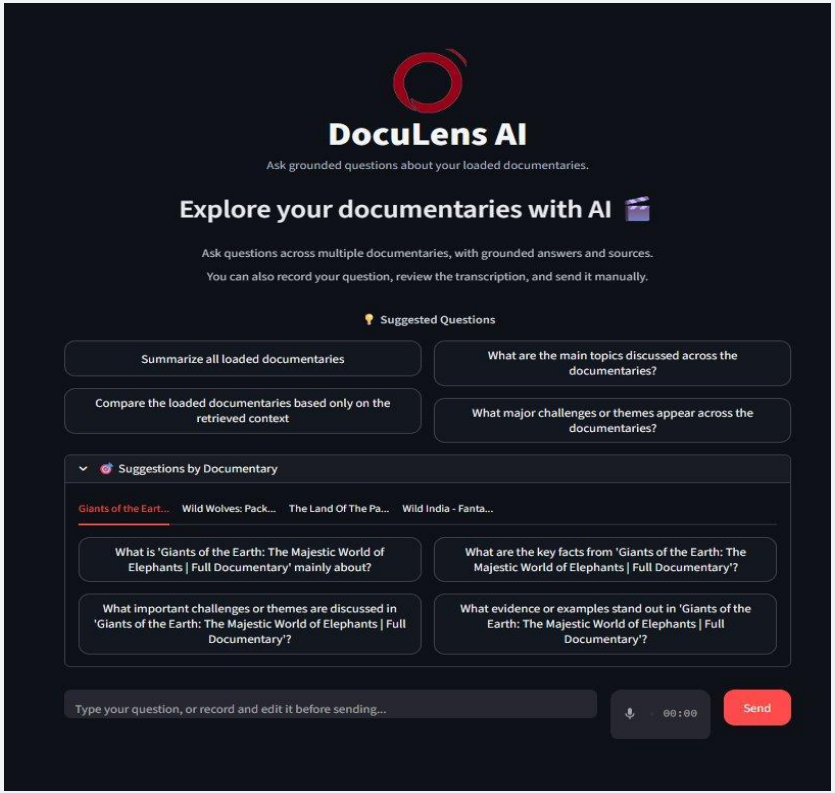
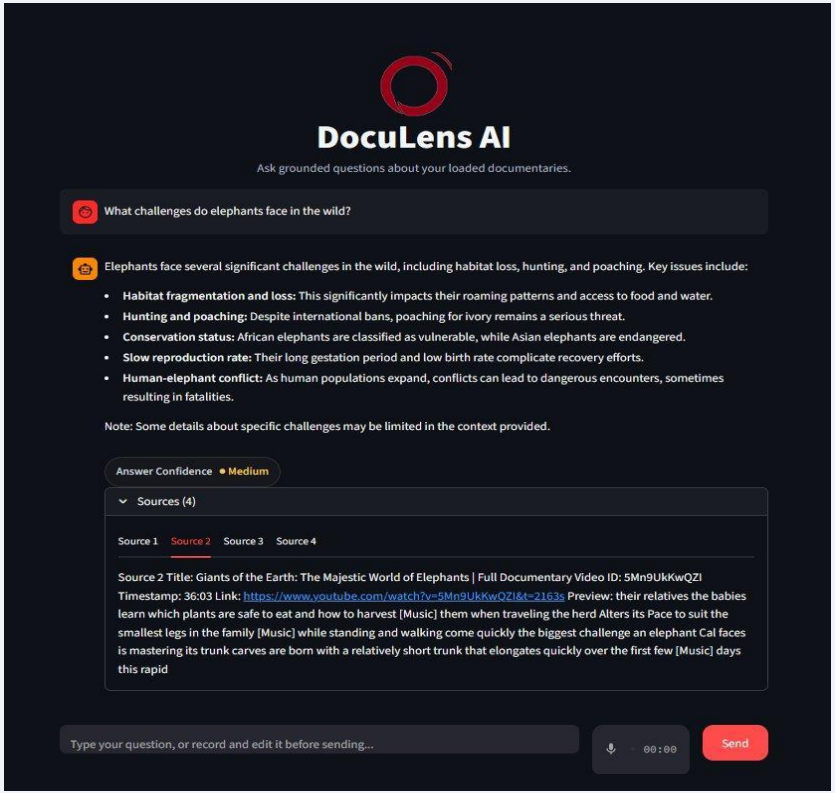
- Full AI pipeline design
- LangGraph agent + tool architecture
- Production monitoring with LangSmith
- Cloud deployment on Streamlit

Future Improvements

- Support non-English documentaries
- Automated evaluation with LangSmith datasets
- Video summarization on load
- User authentication for personal knowledge bases

Demo — DocuLens in Action

App Interface



Q&A with Timestamp Sources